

NFR to Architecture Mapping Matrix

Project: AI-Powered Expense Analyzer & Budget Assistant | Prepared by: Priyam Rai

Architectural Layer	Associated NFR(s)	Key Design Mechanisms & Database Alignments
1. Presentation / API Router Layer	NFR-01 (Performance) NFR-02 (Security) NFR-06 (Scalability)	- Stateless REST API Controllers permit horizontal scaling. - Token validation middlewares (JWT verification router). - Express/fast-route endpoints resolving client operations under 500ms. - Strict input sanitation to prevent cross-site scripting (XSS).
2. Application / Business Logic Services	NFR-07 (Maintainability) NFR-08 (Reliability)	- Modular design patterns separating Controllers, Services, and Repositories. - Comprehensive transactional boundaries ensuring atomic SQL mutations. - Enforce strict error handling routines with custom system-wide error models. - Unit test coverage target framework checking all custom business validation logs.
3. AI / Prediction Service	NFR-04 (Accuracy) NFR-06 (Scalability)	- Decoupled, asynchronous worker process for training models (e.g. Scikit-learn). - Monthly offline retraining cycles to avoid impacting live database queries. - Fail-safe moving average predictive estimation when transaction history is dry. - Sub-second batch predictions queried periodically and cached.
4. Alerting & Notification Engine	NFR-03 (Availability) NFR-08 (Reliability)	- Compares transaction amount inputs against user-defined budget records in D4. - Triggers real-time alerts automatically at 80% (Warning) and 100% (Exceeded). - Offloads notification dispatches to background queues (e.g., SMTP gateways). - Inserts records atomically into the `alerts` database table.
5. Database / Storage Tier (MySQL)	NFR-01 (Performance) NFR-02 (Security) NFR-05 (Data Integrity)	- Exact `DECIMAL(10,2)` columns utilized to avoid rounding discrepancies. - Foreign key constraints enforced with `ON DELETE RESTRICT` (relational integrity). - Hard limits: `CHECK (amount > 0)` and `UNIQUE (user_id, month, year)` constraints. - Indexes on `user_id`, `category_id`, and `expense_date` for sub-second retrieval. - User Soft-Delete status managed via `is_active` flag, protecting operational logs.